

Tanisha Sinha

AI / ML Engineer

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Education

Chhattisgarh Swami Vivekanand Technical University

B.Tech Hons. in CSE (Data Science)

2021 – 2025

CGPA: **8.14**

Skills

ML/AI: Python, Machine Learning, LLM Fine-tuning, Prompt Engineering, NLP, RAG, Agentic AI Scikit-learn, Hugging Face

Engineering: SQL, Git & GitHub, FastAPI, Docker, API Integration

Analytics & BI: Data Visualization, Power BI, Excel

Work Experience

4th Orbit

Jul 2025 – Present

AI/ML Engineer

- Built ML/AI core for two internal products — **Orbithyre** (AI recruitment) and **Gaplytiq** (job readiness) — owning end-to-end model development, NLP pipelines, and LLM integration.
- Developed candidate-job matching engine (scikit-learn) with feature engineering on skills, experience, and role metadata; improved match quality **45%**. Screening pipeline processed **1,000+** profiles, cutting manual review time **60%**.
- Built NLP pipelines for resume parsing, semantic skill extraction, and ATS-optimized resume generation using Hugging Face Transformers and prompt-chained LLM workflows.
- Designed LLM-as-evaluator mock interview system assessing technical depth, role fit, and communication quality with structured feedback generation.
- Engineered skill-gap analysis pipeline comparing competency profiles against JD requirements; produced ranked upskilling recommendations via LangChain reasoning chains.
- Built recruiter-configurable assessment workflows with dynamic, difficulty-targeted question generation powered by LLM pipelines.
- Developed hiring analytics and job-readiness dashboards backed by ML-scored signals and pipeline conversion data.

Tech Stack: Scikit-learn, Pandas, NumPy, Hugging Face Transformers, LangChain, OpenAI API, FastAPI, Node/Express, PostgreSQL, Docker

IIIT Naya Raipur

Jan 2024 – Jun 2024

Research Intern

- Conducted performance analysis of Apache Spark job schedulers for large-scale big data workloads, evaluating **3+** scheduling strategies to optimize resource utilization and execution efficiency.
- Benchmarked experiments on **100K+** record datasets, reducing job completion time **25%** and improving cluster resource utilization **30%** through scheduler tuning.
- Analyzed key metrics (latency, throughput, memory usage, task distribution) to identify bottlenecks and optimization opportunities.
- Published research findings accepted at **TENSYMP 2024**, contributing practical insights to big data analytics and distributed computing.

Tech Stack: Apache Spark, PySpark, Hadoop, YARN, Python, Pandas, NumPy, Jupyter Notebook, Linux

Projects

Powered Data Analytics Platform for Business Intelligence

Apr 2025

- Built a real-time BI platform using Streamlit delivering interactive dashboards monitoring **10+** **KPIs** across operations, sales, and performance.
- Implemented ARIMA-based forecasting achieving **20%+** prediction accuracy improvement for trend and demand forecasting.
- Developed Isolation Forest anomaly detection with **90%+** detection efficiency, enabling faster risk mitigation.
- Integrated automated email alerting (Low/Medium/High severity), reducing incident response time **60%**.

Tech Stack: Python, Streamlit, ARIMA, Isolation Forest, Scikit-learn, smtplib

Predictive Maintenance for Industrial Machinery

Jan 2025

- Developed Streamlit-based predictive maintenance app analyzing historical sensor data to forecast machinery failures before breakdowns.
- Trained ML models on **100K+** sensor records, achieving **88%+** prediction accuracy for early failure detection.
- Reduced unplanned downtime risk **35%**; lowered projected operational repair costs **20%+** through proactive recommendations.
- Integrated MySQL for structured storage; built interactive dashboards visualizing health trends, failure probabilities, and maintenance schedules.

Tech Stack: Python, MySQL, Scikit-Learn, Streamlit, Matplotlib, Pandas, NumPy, Seaborn

Performance Analysis of Apache Spark Job Schedulers

Jun 2024

- Comparative analysis of Spark schedulers (FIFO, Fair, Capacity) across **3+** workload scenarios; reduced average job execution time **25%**.
- Improved cluster resource utilization **30%** by identifying optimal scheduler for multi-tenant and batch workloads.
- Processed **100K+** record datasets using PySpark, simulating real-world big data environments.

Tech Stack: Hadoop, Apache Spark, PySpark, Big Data Analytics, Performance Benchmarking

Achievements

- Paper on Apache Spark job scheduler performance analysis accepted at **TENSYMP 2024**.
- Event Coordinator at TPO — oversaw placement drives and career development events ensuring seamless execution.